填空

1.

$$a=\frac{dv}{dt}=\frac{dv}{dt}\cdot\frac{dx}{dx}=v\frac{dv}{dx}=3+2x$$

$$\int_5^v vdv=\int_0^3(3+2x)dx \qquad v_3=\sqrt{61}=7.81m/s$$

2.

$$v=\omega R=Rkt^2 \qquad k=\frac{v}{Rt^2}=4$$

$$a_n=\omega^2R=Rk^2t^4=2\mathrm{m/s^2} \qquad a_\tau=\alpha R=R\frac{\mathrm{d}\omega}{\mathrm{d}t}=2Rkt=8\mathrm{m/s^2}$$

3.

$$\vec{v}=\frac{d\vec{r}}{dt}=\frac{10}{3}t\vec{i}$$

$$A=\frac{1}{2}mv_2^2-\frac{1}{2}mv_1^2=2.2J$$

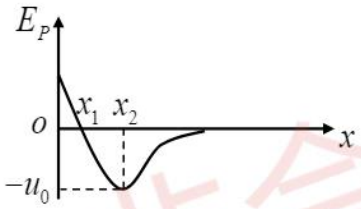
4.

$$E_P=\int_{3R}^{R}-\frac{GMm}{r^2}dr=\frac{2GMm}{3R} \qquad E_P=\int_{3R}^{\infty}-\frac{GMm}{r^2}dr=-\frac{GMm}{3R}$$

5.

$$\int_0^1 kxdx=\int_1^x kxdx \qquad x=\sqrt{2} \qquad \Delta x=x-1=0.41cm$$

6.



$$x \geq x_1$$

$$E_k = u_0$$

7.

$$F = mg + ma = mg + m(3 + 5t)$$

$$\int_0^2 F dt = 356N \cdot S \qquad \Delta P = \int_0^2 mad t = 160N \cdot S$$

8.

$$m\frac{d\vec{v}}{dt} = \vec{F} + (\vec{u} - \vec{v})\frac{dm}{dt} \qquad a = \frac{F}{M - \Delta mt}$$

$$\frac{dv}{dt} = \frac{F}{M - \Delta mt} \qquad \int_0^v dv = \int_0^t \frac{F}{M - \Delta mt} dt \qquad v = \frac{F}{\Delta m} Ln \frac{M}{M - \Delta mt}$$

9.

$$\Omega = \frac{M}{L \sin\theta} = \frac{Wh}{J_0 \omega}$$

方向：从上往下逆时针方向

10.

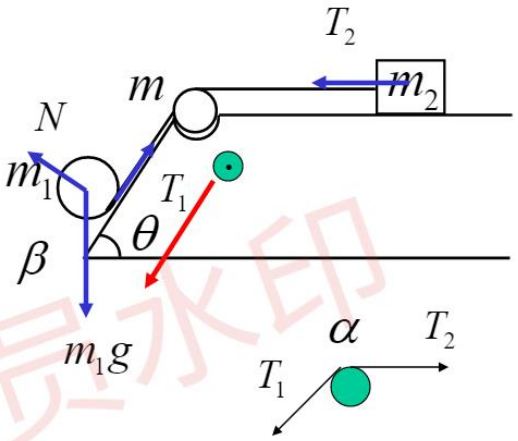
$$\sin\theta = \frac{a}{\sqrt{x^2 + a^2}}$$

$$\cos\theta \frac{d\theta}{dt} = \frac{ax}{(x^2 + a^2)^{3/2}} \frac{dx}{dt} \qquad \omega = \frac{av}{x^2 + a^2}$$

$$E_k = \frac{1}{2} J\omega^2 = \frac{mL^2 a^2 v^2}{6(x^2 + a^2)^2}$$

二、计算题

1、



$$m_1 g\sin\theta - T_1 = m_1 a_c$$

$$T_1 R_1 = J_c\beta \qquad J_c = \frac{1}{2}m_1 R_1^{\ 2}$$

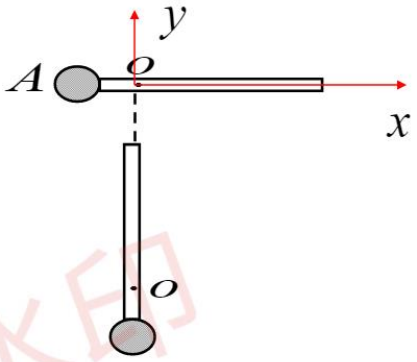
$$rT_1 - rT_2 = mr^2\alpha$$

$$T_2 = m_2 a_2$$

$$a_2 = r\alpha$$

$$a_2 = a_c - \beta\, R_1$$

3、



(1) $$J_0 = \frac{1}{12}ml^2 + m(\frac{l}{4})^2 + 3m(\frac{l}{4})^2 = \frac{1}{3}ml^2$$

$$M = J\alpha$$

$$3mg\frac{l}{4} - mg\frac{l}{4} = J_0\alpha \qquad \alpha = \frac{3g}{2l}$$

(2) $$x_c = \frac{-3m\frac{l}{4} + m\frac{l}{4}}{4m} = -\frac{1}{8}l \qquad R_c = \frac{1}{8}l$$

$$F_y - 4mg = -4ma_{ct} = -4m\alpha R_c \qquad F_y = \frac{13}{4}mg$$

$$F_x = 4ma_{cn} \qquad F_x = 0 \qquad \vec{F} = \frac{13}{4}mg\vec{j}$$

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(3) 机械能守恒 $4mgR_c = \frac{1}{2}J_0\omega^2$

$$-3mg\frac{l}{4}+mg\frac{l}{4}=\frac{1}{2}J_0\omega^2 \qquad \omega=\sqrt{\frac{3g}{l}}$$

(4) 角动量守恒

$$m_0 v\frac{l}{4}+J_0\omega=\left[m_0(\frac{l}{4})^2+\frac{1}{3}ml^2\right]\omega'$$

$$\omega'=\frac{4v+16\sqrt{3gl}}{17l}$$

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